fml (2).pdf

by Mithil S

Submission date: 30-Oct-2026 12:52AM (UTC+0800)
Submission ID: 3074349575
File name: fml_2_.pdf (496.75K)
Word count: 7370
Character count: 39799

Evaluating Value-Based Escalation for Reliability Monitoring Under Distribution Shift

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***Abstract*—Machine learning models can become unreliable when deployment data differ from the data used during training. Additional runtime monitoring can help identify unreliable predictions, but applying a computationally expensive monitor to every input increases inference cost and latency. Value-based escalation addresses this by prioritizing the inputs for which additional monitoring is expected to give the greatest benefit. The reliability of a learned escalation-value ranking under distribution shift, however, requires empirical evaluation.**

This study investigates whether an escalation-value estimator fitted on clean data can still identify inputs that benefit from additional monitoring when the input distribution changes. The approach uses confidence, predictive entropy and classification margin as inexpensive features for a Random Forest regressor that estimates the benefit of invoking an expensive, disagreement-based monitor. We evaluate it with pretrained CIFAR-10 classifiers on CIFAR-10-C Gaussian noise at five severity levels. This is a pilot-scale evaluation: one corruption family, one data-split seed and one primary budget. At a 20% escalation budget we compare the estimator with confidence-, entropy- and margin-based selection, random selection, cheap-only and always-on monitoring, matched direct abstention, and an entropy-only ablation, using a paired bootstrap.

In this setting the estimator did not outperform simple confidence-based selection. Its mean utility gain over cheap-only prediction was 0.0455, compared with 0.0548 for low confidence. The paired difference was -0.0094 (95% bootstrap CI -0.0120 to -0.0069). The entropy-only ablation performed almost identically to the three-feature estimator (0.0449). An oracle selecting the same number of inputs reached 0.1609; this reference uses ground-truth information unavailable at deployment, but it indicates substantial headroom that the cheap features did not capture. Selective monitoring at 20% required about one third of the latency of always-on monitoring (0.376 versus 1.102 ms per image). Thresholds calibrated on clean data escalated 61% to 78% of shifted inputs instead of 20%. The conclusions apply only to the evaluated pretrained models, monitor rule, corruption family and split seed.

***Index Terms*—distribution shift, runtime monitoring, selective escalation, CIFAR-10-C, uncertainty estimation, model disagreement**

I. Introduction

Machine learning classifiers are increasingly deployed in environments where the distribution of incoming data differs from the distribution seen during training. Changes in acquisition conditions, image quality and operating environment can alter model predictions and reduce classification reliability. This problem, known as *distribution shift*, makes it important to assess not only the predictions of a deployed model but also the conditions under which those predictions can be trusted. In image classification, common corruptions such as noise, blur and digital distortion provide a controlled setting for studying this problem [1]. Applying additional reliability checks to every input, however, adds computational overhead and creates a trade-off between reliability and inference efficiency.

Runtime monitoring is one way to address this challenge. It identifies predictions or operating conditions that may require further scrutiny. Some approaches use predictive disagreement to detect model failures without ground-truth labels, and sequential methods detect changes in model performance over time [2], [3]. The cost of monitoring depends on the method. Maximum softmax confidence, predictive entropy and the margin between the two highest class probabilities come directly from a model's existing output. More expensive checks, such as evaluating additional classifiers and comparing their predictions, require further inference. Invoking an expensive monitor for every input may therefore be inefficient, while relying only on inexpensive signals may miss inputs that would benefit from additional checking.

Cost-aware monitoring and value-based escalation offer two ways to allocate this computation. Cost-constrained runtime monitoring asks which checks to perform under a limited budget [4]. Value-based approaches estimate the benefit of an additional computation or of consulting a stronger decision maker [5], [6]. The distinction matters because uncertainty does not necessarily mean that additional monitoring will improve the final decision. An uncertain input may remain unresolved after a secondary check, and a confidently incorrect prediction may escape a simple uncertainty threshold. A value-based policy therefore ranks inputs by the expected usefulness of monitoring rather than by uncertainty alone.

A learned value estimator raises a further question. Its predictions depend on the relationship between inexpensive signals and the benefit of monitoring in the data used for fitting. When the deployment distribution changes, that relationship may change too. An estimator that ranks inputs well on clean data may then give high priority to inputs that gain little from escalation and overlook inputs that would gain. The related literature addresses value-aware computation, cost-constrained monitoring and deployment reliability. The present study asks a narrower empirical question: does a value-of-

escalation ranking learned on clean image data remain useful as corruption severity increases? This gap is framed with respect to the literature reviewed here. We do not claim that the robustness of value-based policies under shift has never been studied.

To examine this question, we evaluate a selective monitoring pipeline using CIFAR-10 [7] as the clean source and CIFAR-10-C [1] as the controlled-shift benchmark. The pipeline extracts confidence, entropy and margin from a base classifier. A Random Forest regressor [8] uses them to estimate the benefit of invoking a disagreement-based monitor. The regressor is fitted on offline targets defined as the utility difference between monitored and unmonitored decisions. At evaluation time, escalation depends only on the inexpensive features, not on the oracle targets. We compare the approach with uncertainty-based policies, a cheap-only policy, an always-on policy and a matched direct-abstention policy under identical escalation budgets. The experiments reported here are pilot-scale: they cover Gaussian noise at five severities, a single split seed and a 20% primary budget. Extending them to the remaining corruption families, split seeds and budgets is left to future work.

This work makes three contributions.

1) **Problem evaluation.** We examine whether an escalation-value estimator fitted on clean data can rank inputs by the benefit of additional monitoring under Gaussian-noise shift of increasing severity.
2) **Policy comparison.** We compare value-based escalation with uncertainty-based and other selection policies under matched budgets, using utility, ranking agreement and a paired bootstrap analysis.
3) **Cost–reliability analysis.** We relate monitoring allocation, predictive outcomes, threshold behaviour and inference overhead across severities.

These contributions concern the empirical evaluation of a specified method in a specified setting. They do not imply that value-based escalation, uncertainty estimation or runtime monitoring is itself new.

The rest of the paper is organized as follows. Section II reviews related work. Section III formulates the problem and describes the method. Section IV describes the data, baselines, metrics and protocol. Section V reports the results. Section VI discusses them and states the limitations, and Section VII concludes.

II. Related Work

Research on selective reliability monitoring under distribution shift intersects four areas: deployment-time monitoring, cost-constrained monitor selection, value-based escalation and uncertainty-based decision policies. Deployment monitoring seeks to identify changes or failures in model behaviour. Cost-aware methods allocate limited monitoring resources. Value-based escalation tries to identify inputs for which additional computation is beneficial. The present study connects these areas by testing whether a learned escalation-value ranking stays useful when the deployment distribution differs from the clean data used to fit it.

Deployment-time reliability monitoring. A central challenge in deployed machine learning is to distinguish shifts that degrade predictive performance from shifts that do not require intervention. Nguyen et al. [2] propose a disagreement-based approach for detecting post-deployment model failures without labels, which establishes predictive disagreement as a practical reliability signal. Prinster et al. [3] propose WATCH, a sequential monitoring framework based on weighted conformal martingales that supports online monitoring under changing distributions while controlling false alarms. Zhang et al. [9] study prediction-powered risk monitoring, which combines model predictions with a limited number of true labels to monitor deployed risk. These methods mainly address the detection or statistical monitoring of degradation. The present study instead concerns the instance-level allocation of an expensive monitor: given a budget, which inputs are most likely to benefit from invoking it?

Selective prediction and calibration. Selective classification lets a classifier abstain on inputs where its prediction is likely to be wrong, trading coverage for risk [6]. Confidence-based rejection depends on how well confidence reflects correctness, and modern networks are often poorly calibrated [10]. These works motivate the confidence, entropy and margin baselines used here, and the direct-abstention baseline that abstains on the least confident inputs at matched coverage.

Cost-constrained runtime monitoring. Monitoring decisions must account for the computational cost of additional checks. Hua et al. [4] formulate the combination of runtime monitors as a budget-constrained optimization problem, in which the allocation of monitor calls and interventions under cost constraints affects the balance between detection performance and resource use. This is directly relevant to selective escalation, where invoking a monitor on every input may be impractical. Selecting a protocol from known monitor costs differs, however, from learning an instance-level estimate of the benefit of escalation and testing it under distribution shift.

Value-based escalation. Uncertainty signals are inexpensive, but they do not directly measure whether a secondary model or monitor will improve the decision. Pona et al. [5] address this distinction in *Calibrate-Then-Delegate*, which studies safety monitoring for language-model cascades. They learn a delegation-value probe that estimates the benefit of escalating an input to a more expensive expert, and they calibrate a threshold to control cost. This work is a close conceptual precedent because it treats the value of escalation as distinct from uncertainty; it does not study image classification. We therefore do not claim that value-based escalation is a new idea. Our study differs in its setting and focus: image classification under a controlled shift from clean to corrupted data, a disagreement-based monitor with abstention, a gate fitted only on clean data, and an analysis of ranking quality, utility, threshold behaviour and latency across corruption severities. We make no claim of shift guarantees.

Positioning. The reviewed literature provides methods for detecting deployment deterioration, monitoring risk, selecting cost-constrained checks and predicting the benefit of additional computation. The question investigated here is narrower. When a value estimator is fitted on clean CIFAR-10 using inexpensive predictive signals, does its ranking remain informative on CIFAR-10-C as severity increases? Table I summarizes how the closest works relate to this study.

III. Methodology

A. Problem Setting

Consider a deployed image classifier f that maps an input image $x \in \mathcal{X}$ to a probability vector over K classes and predicts the class with the highest probability. Let P_s be the source distribution and P_t a target distribution met during deployment. Clean CIFAR-10 images represent P_s, and CIFAR-10-C corruption conditions represent target distributions that vary by corruption type and severity. The classifier and the escalation-value estimator are kept fixed during shifted evaluation, so changes in performance can be examined without retraining on corrupted data.

The system has two levels of computation. The first consists of inexpensive signals derived from the classifier's output. The second is an expensive monitor that requires additional model inference. For each input the system decides whether to use only the base prediction or to invoke the monitor. The objective is to spend a limited monitoring budget on the inputs for which the additional computation is expected to help most.

B. Base Classifier and Cheap Features

For an input x, the base classifier produces $p(x) = (p_1(x), \ldots, p_K(x))$ with $\sum_k p_k(x) = 1$. Let $p_{(1)}(x)$ and $p_{(2)}(x)$ be the largest and second-largest probabilities. Three features are extracted:

$$c(x) = \max_k p_k(x), \tag{1}$$

$$H(x) = -\sum_{k=1}^{K} p_k(x) \log p_k(x), \tag{2}$$

$$m(x) = p_{(1)}(x) - p_{(2)}(x). \tag{3}$$

Here c is the probability of the predicted class, H measures how dispersed the distribution is (with the convention $0 \log 0 = 0$), and m measures how close the two leading classes are. The feature vector is $z(x) = [c(x), H(x), m(x)]^\top \in \mathbb{R}^3$. These features come from the base classifier's existing output and do not require running the monitor.

C. Disagreement-Based Monitor

The monitor consists of $J = 3$ additional classifiers with predictions $h_1(x), h_2(x), h_3(x)$. Let $\hat{y}_f(x)$ be the base prediction. The acceptance rule is

$$A(x) = \begin{cases} 1, & \sum_{j=1}^{3} \mathbb{1}[h_j(x) = \hat{y}_f(x)] \geq 2, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

When $A(x) = 1$ the base prediction is served. When $A(x) = 0$ the system abstains. The monitor therefore never changes the predicted class: it only decides whether the base prediction is served or withheld, and it does not correct errors. The threshold of two agreeing monitors was fixed before any results were seen and was not tuned. The rule is an operational reliability check, not a guarantee of correctness, because models can agree on a wrong answer.

D. Utility and Oracle Escalation Value

Let $a_0(x)$ be the action without monitoring (serve the base prediction) and $a_1(x)$ the action after monitoring (serve the base prediction if $A(x) = 1$, otherwise abstain). Let $U(a, y)$ be the utility of an action when the true class is y. We use

$$u_0 = U(a_0(x), y) = \begin{cases} +1 & \text{base prediction correct,} \\ -1 & \text{base prediction wrong,} \end{cases} \tag{5}$$

and $u_1 = U(a_1(x), y) = u_0$ if the prediction is accepted and $u_1 = 0$ if the system abstains. The oracle escalation value is

$$V^*(x, y) = u_1 - u_0 \in \{-1, 0, +1\}. \tag{6}$$

Three cases follow. A value of $+1$ means the monitor abstained on an input where the base prediction was wrong (useful escalation). A value of -1 means the monitor abstained on an input where the base prediction was correct, withholding a correct answer (harmful escalation). A value of 0 means the monitor accepted the base prediction, so utility did not change (wasted escalation). The oracle value needs the ground-truth label, so it is a retrospective reference used only offline to build training targets and to evaluate policies. It is never available to the deployed policy and is not a practical method. Uncertainty is not the same as escalation value: confidence, entropy and margin describe the base output, whereas V^* measures the actual utility change caused by the monitor.

E. Learned Value Estimator

A Random Forest regressor $g : \mathbb{R}^3 \to \mathbb{R}$ (200 trees, minimum leaf size 10) is fitted on the gate-fit split to predict V^* from $z(x)$:

$$\hat{V}(x) = g(z(x)). \tag{7}$$

At deployment the regressor receives neither the label nor the monitor output. As an ablation, a second regressor with the same hyperparameters is fitted using only the entropy feature $H(x)$. Comparing the two shows whether the three-feature representation adds value.

F. Budget-Constrained Escalation Policies

Let $e_i \in \{0, 1\}$ indicate whether input x_i is escalated. For a batch of N inputs and budget b,

$$\sum_{i=1}^{N} e_i \leq \lfloor bN \rfloor, \tag{8}$$

and the policy escalates the highest-scoring inputs up to that limit. Ties are broken deterministically by index. The utility

TABLE I
COMPARISON OF THE CLOSEST RELATED WORKS WITH THE PRESENT STUDY.

Work	Main idea	Relation to this study	Difference
Nguyen et al. [2]	Disagreement-based detection of deployment failures without labels	Motivates the disagreement-based monitor	Detects failure; does not allocate a monitor per input under a budget
Prinster et al. [3]	Sequential monitoring with weighted conformal martingales	Deployment monitoring under shift	Statistical alarms over time, not instance-level escalation
Zhang et al. [9]	Prediction-powered risk monitoring	Monitoring of deployed risk	Estimates risk; no escalation policy
Hua et al. [4]	Budget-constrained selection of runtime monitors	Cost-aware monitoring	Selects protocols from known monitor costs; no learned per-input value
Pona et al. [5]	Learned delegation-value probe with calibrated threshold (language-model cascades)	Closest conceptual precedent for value-based escalation	Not studied for a clean-fitted gate on CIFAR-10-C with an abstaining disagreement monitor
Geifman and El-Yaniv [6]	Selective classification with abstention	Source of the abstention and coverage framing	Abstains on uncertainty; does not model monitor benefit

of a policy on input i is $u_{0,i} + e_i V_i^*$. The following policies are evaluated:

- **Value gate:** ranks inputs by $\hat{V}(x)$.
- **Value gate (entropy-only):** the ablation above.
- **Low confidence, high entropy, low margin:** rank by $-c(x)$, $H(x)$ and $-m(x)$.
- **Random:** selects inputs uniformly at random, averaged over 200 draws.
- **Oracle:** ranks by true V^*. It is the best fixed-budget escalation set for this monitor under the defined utility, but it is not deployable and is not an upper bound for direct abstention.
- **Cheap-only:** never invokes the monitor.
- **Always-on:** invokes the monitor for every input.
- **Direct abstention (matched):** abstains on the least confident images, with the number of abstentions set equal to those of the value gate, so that coverage is matched. It uses no monitor.
- **Fixed-threshold variants:** the value gate and the confidence rule with thresholds calibrated on clean data to meet the 20% budget, then applied unchanged to shifted data. Their actual escalation rates may drift under shift.

G. Statistical Analysis

The primary endpoint of this analysis is the mean per-image utility difference between the value gate and the low-confidence policy at the 20% budget. It is computed as the macro-average, over the five Gaussian-noise conditions, of the condition-specific mean differences. Because the five conditions share the same evaluation images, this equals the pooled per-image difference. Because u_0 cancels in a paired difference, the per-image difference is $V_i^* (e_i^{\text{gate}} - e_i^{\text{conf}})$. Throughout, every aggregate metric in the results is a macro-average over the five corrupted conditions.

Confidence intervals use a paired bootstrap with 2000 replicates, resampling evaluation-image indices jointly across the five conditions with the selections held fixed. This preserves the pairing between corrupted versions of the same image and makes the interval conditional on the fitted gate and the chosen selections. It does not capture variability from refitting the gate or from the random baseline's draws. All comparisons other than the primary endpoint are secondary or exploratory. The following decision rules were used, based on the sign of the primary difference and whether its confidence interval excludes zero:

1) *Gate beats confidence:* the primary difference is positive and its confidence interval excludes zero.
2) *Confidence beats gate:* the primary difference is negative and its interval excludes zero.
3) *Gate versus direct abstention:* utility compared at matched coverage, reported with the number of monitor calls used.
4) Otherwise the outcome is reported as mixed or no advantage. An interval that includes zero is not treated as proof of equivalence.

The primary comparison was evaluated on a single data split (seed 42). The resulting interval describes uncertainty under the stated image-resampling procedure, conditional on the fitted gate. It does not establish robustness across independent random splits.

IV. EXPERIMENTAL SETUP

A. Datasets

CIFAR-10. CIFAR-10 [7] contains 60,000 colour images of size 32×32 in ten balanced classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), divided into 50,000 training and 10,000 test images.

CIFAR-10-C. CIFAR-10-C [1] applies common corruptions to the 10,000 test images at five severity levels, with the original labels retained. The executed experiment uses *Gaussian noise* at severities 1–5. Together with the clean set this gives

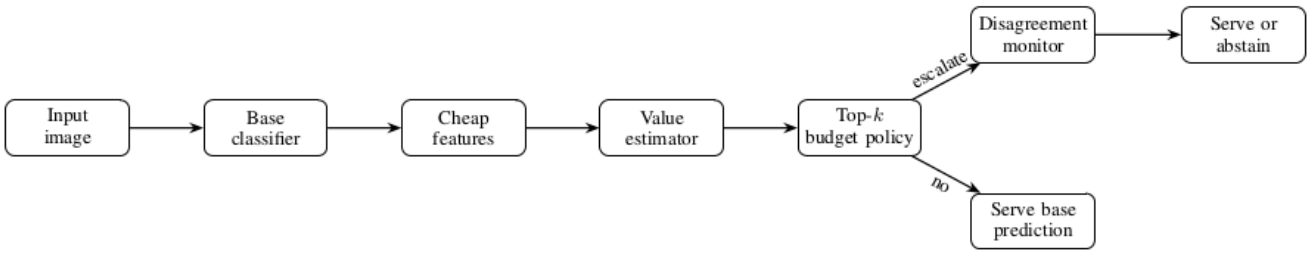


Fig. 1. Selective reliability-monitoring pipeline. The base classifier produces a prediction and inexpensive features. The value estimator ranks inputs by predicted escalation benefit, and the policy allocates the monitor under a fixed budget. The monitor either accepts the base prediction or causes abstention. Oracle values are computed offline and are not available to the deployed policy.

six evaluation conditions (one clean and five corrupted). The remaining families planned in the synopsis (defocus blur, frost and JPEG compression) were not evaluated. The data were obtained from the Zenodo release, keeping only the Gaussian-noise file and the label file. Before any analysis, we verified that every severity block carries the clean test labels in the same order.

Preprocessing. Images are scaled to $[0, 1]$ and normalized with channel means $(0.4914, 0.4822, 0.4465)$ and standard deviations $(0.2471, 0.2435, 0.2616)$, as required by the selected checkpoints. The same preprocessing is applied to clean and corrupted images.

B. Models

We use publicly available pretrained CIFAR-10 classifiers [11]: ResNet-18 as the base classifier and ResNet-34, ResNet-50 and DenseNet-121 as the three monitor classifiers. *We did not train these networks.* This deviates from the original synopsis, which proposed training a ResNet-18 and an ensemble. The deviation keeps the research question and the rest of the method intact while reducing compute. The repository's primary weight-download link (a Box URL) returned HTTP 404 at run time, so the weights were obtained from the repository's Google Drive backup. An obsolete `pytorch_lightning` import in the repository's `module.py` was commented out because inference does not use it. As a correctness check, clean accuracy on all 10,000 test images must match the published accuracies within 0.5 percentage points, and the run stops otherwise.

C. Data Splits

The pretrained networks have already seen all 50,000 CIFAR-10 training images, so no training-set holdout is available for fitting the gate without leakage through the base model. For the split seed used (42), the 10,000 clean test images are therefore partitioned into three disjoint subsets: 1,500 images to fit the gate, 500 to calibrate fixed thresholds, and 8,000 for evaluation. The gate is fitted on clean data only. All six conditions are evaluated on the same 8,000 evaluation indices, so images and labels are aligned across conditions, and the gate-fit images are never used for evaluation. The split indices are saved. Because all three subsets come from the official test set, this is a pilot protocol and not a conventional untouched-test evaluation; Section VI-E discusses the consequences.

D. Baselines and Budget

The policies described in Section III serve as baselines. Selective policies are compared at a matched escalation budget of 20% of the 8,000 evaluation images (1,600 escalated images), which is the primary budget. Matching the budget prevents a policy from appearing better only because it invokes the monitor more often. The code also supports 10% and 30% budgets, but results for these budgets were not available in the saved outputs and are not reported.

E. Evaluation Metrics

Utility. Mean per-image utility under the $+1/-1/0$ definition above, and the utility gain over the cheap-only policy. For a single condition, utility equals $C(1-2R)$, where C is coverage and R is the error rate among served predictions. All reported aggregates are macro-averages over the five corrupted conditions, so the aggregate utility need not equal $\bar{C}(1-2\bar{R})$ computed from the averaged coverage and error rate.

Spearman rank correlation. The agreement between predicted values $\hat{V}$ (or the baseline scores) and the oracle values V^* over all evaluation images within each condition. It does not depend on the escalation budget. Because V^* is mostly zero with heavy ties, Spearman correlation is always read alongside utility and never on its own.

Risk and coverage. Coverage is the fraction of images for which a prediction is served. The error rate among served predictions is the fraction of wrong served predictions, reported together with coverage so that abstention is not mistaken for improvement.

Escalation rate and monitor invocations. The achieved escalation rate and the average number of monitor-model calls per image (escalation rate times three).

Latency. Mean and 95th-percentile batch execution time, measured separately for each model, the gate and the complete selective pipeline, on one machine at one batch size. Timing includes a warm-up call and seven repetitions on 2,000 clean images with GPU synchronization. With only seven repetitions the 95th percentile is unstable, and it describes batch time, not the tail latency of individual requests.

TABLE II
CLEAN TEST ACCURACY OF THE PRETRAINED MODELS (%).

Model	Published	Measured
ResNet-18 (base)	93.07	93.07
ResNet-34 (monitor)	93.34	93.33
ResNet-50 (monitor)	93.65	93.65
DenseNet-121 (monitor)	94.06	94.06

F. Experimental Protocol

The protocol follows a fixed sequence. (1) Load the models and check clean accuracy. (2) Run all models on all six conditions and cache the probability outputs. (3) Compute the oracle values, fit the gate and the entropy-only gate on the gate-fit split, and calibrate fixed thresholds on the calibration split. (4) Freeze the gate and policies, then evaluate every condition on the evaluation split. (5) Run the paired bootstrap and the primary-endpoint analysis. (6) Run the integrity checks and the latency benchmark. The integrity checks confirm that the splits partition the test set, that policy escalation rates match their budgets, that the oracle dominates the gate among escalation policies, and that the bootstrap point estimate equals the difference in mean utility gains. The execution log reports that all checks passed.

Only one split seed (42) was run. It changes the gate and the data partition, but the pretrained network weights stay fixed.

G. Computing Environment

Experiments ran in a GPU-enabled Kaggle notebook using Python 3.13, PyTorch [12] 2.11.0 (CUDA 12.8), scikit-learn [13], NumPy and Matplotlib, with batch size 256. The notebook's total runtime was 53.5 minutes, of which about 1.1 minutes was model inference. Most of the remainder was spent downloading data.

V. RESULTS

All values come from the saved outputs of the executed run: Gaussian noise at five severities, one split seed, and a 20% budget. Analyses for other budgets, per-condition intervals and the clean-condition ranking correlation were not available in the saved outputs and are not reported.

A. Clean Accuracy and Value Targets

Table II reports clean test accuracy on all 10,000 images. All models matched their published accuracy within about 0.01 percentage points.

On the 1,500-image gate-fit split, $V^* = 0$ for 1,430 images (95.3%), $V^* = +1$ for 46 (3.1%) and $V^* = -1$ for 24 (1.6%). The targets are therefore sparse: only 70 images carry a non-zero value. The fitted gate's feature importances were 0.398 for entropy, 0.339 for confidence and 0.263 for margin.

Across the five corrupted conditions, the base classifier's mean error rate was 41.9% (cheap-only utility 0.162).

TABLE III
SPEARMAN CORRELATION BETWEEN VALUE-GATE SCORES AND V^* UNDER GAUSSIAN NOISE (SEED 42; VALUES READ FROM FIG. 2).

Severity	1	2	3	4	5
Gate vs. V^*	0.109	0.132	0.138	0.129	0.131

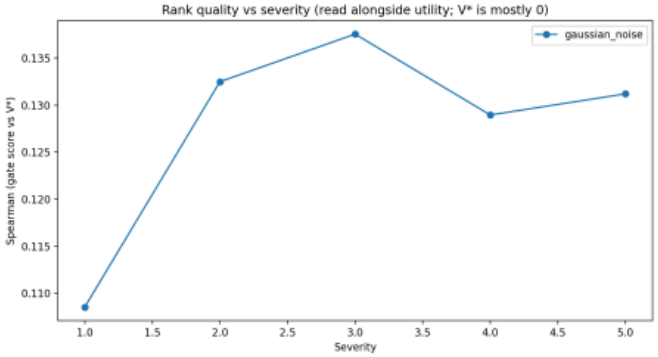


Fig. 2. Spearman correlation between value-gate scores and oracle values V^* across Gaussian-noise severities (seed 42).

B. Value-Ranking Performance Under Gaussian Noise

Table III gives the Spearman correlation between the value-gate score and V^* at each severity, computed over all evaluation images. The values were read from Fig. 2 and should be regenerated at full precision from the saved result files. The mean over the five severities was 0.128. Correlations were positive at every severity and did not decrease monotonically with severity. The clean-condition correlation was not available in the saved outputs.

C. Primary Endpoint and Policy Comparison

Table IV reports the primary endpoint: the mean utility difference between the value gate and low-confidence escalation at the 20% budget, macro-averaged over the five corrupted conditions.

The difference is negative and its interval excludes zero. Under the decision rule above, the analysis returns the verdict "confidence beats gate" for this seed. The difference between the rounded policy means in Table V is -0.0093; the reported -0.0094 was computed from unrounded values. Because only one split seed was run, the result is not evidence of consistency across partitions, and the interval is conditional on the fitted gate.

Table V compares all policies at the 20% budget, averaged over the five corrupted conditions.

All three uncertainty-based rules gained between 0.0534 and 0.0548, which exceeds the gate's 0.0455 and the entropy-only gate's 0.0449. Random selection gained 0.0173. The oracle gained 0.1609, nearly three times the best deployable escalation rule, so the cheap features left most of the achievable benefit unused. The oracle uses ground-truth information unavailable at deployment. Its gain also exceeded that of always-on monitoring (0.0861), because the monitor

TABLE IV
PRIMARY ENDPOINT: VALUE GATE MINUS LOW CONFIDENCE AT THE 20% BUDGET, AVERAGED OVER FIVE GAUSSIAN-NOISE CONDITIONS.

Split seed	Difference	95% CI
42	−0.0094	[−0.0120, −0.0069]

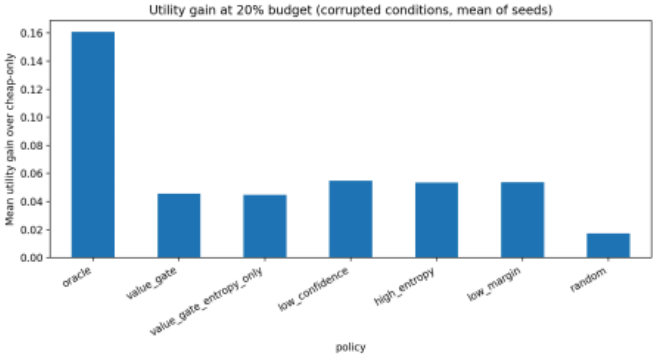


Fig. 3. Mean utility gain over cheap-only at the 20% budget (corrupted conditions, seed 42). The oracle is an offline reference.

sometimes withholds a correct answer ($V^* = -1$) and the oracle avoids those cases.

Relative to always-on monitoring, the value gate achieved about 53% of the gain (0.0455 of 0.0861) with 20% of the monitor calls, and low confidence achieved about 64% (0.0548 of 0.0861).

D. Comparison With Direct Abstention

At matched coverage (0.8981), the value gate achieved a utility of 0.2075 and direct abstention 0.2046, a difference of 0.0029 in means. The gate used 0.6 monitor calls per image, while direct abstention used none. No bootstrap interval for this difference is available in the saved outputs, so the small difference is descriptive only.

E. Ablation: Three Features Versus Entropy Only

Adding confidence and margin changed the mean utility gain by 0.0006 and raised the mean Spearman correlation from 0.109 to 0.128. The utility difference is small, no interval is available for it, and it is not evidence that the extra features are beneficial.

F. Fixed-Threshold Behaviour Under Shift

Thresholds calibrated on clean data to give a 20% escalation rate escalated far more inputs under noise. Averaged over the five severities, the achieved rate was 0.783 for the value gate and 0.608 for the confidence rule (Table V). Fig. 4 shows the trend by severity. For the confidence rule the rate rose from roughly 0.34 at severity 1 to roughly 0.77 at severity 5, as confidence fell under stronger noise. For the value gate the rate started near 0.87 at severity 1 and decreased to near 0.73 at severity 5. These severity-level values are read from the figure. Because these rules escalate several times more inputs

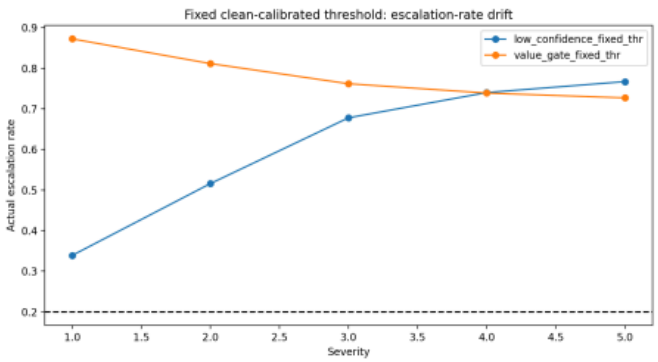


Fig. 4. Actual escalation rate of clean-calibrated fixed thresholds versus Gaussian-noise severity. The dashed line marks the nominal 20% budget.

than the nominal budget, their higher utility gains (0.0871 and 0.0674) are not comparable with the budget-matched policies.

G. Latency and Computational Overhead

The selective pipeline took 0.376 ms per image, about 2.5 times the cheap-only pipeline (0.151 ms) and about one third of the always-on pipeline (1.102 ms). The gate itself added 0.035 ms per image, which is about 23% of the base classifier's time. The selective timing adds the gate and the monitor calls for the selected 20% of images.

H. Summary of Findings

In this pilot-scale setting, the value gate improved on random selection and had a slightly higher mean utility than matched direct abstention (without an interval), but it did not match simple uncertainty rules. Its utility gain at the 20% budget was 0.0455, against 0.0548 for low confidence, and the paired difference of −0.0094 had an interval that excluded zero for this split seed. The three-feature estimator was almost indistinguishable from the entropy-only estimator in utility. The oracle gain was 0.1609, which indicates large unused headroom. Clean-calibrated thresholds did not hold the nominal rate under shift. Selective monitoring cost about one third of the always-on batch latency. These findings are limited to Gaussian noise, one split seed, one primary budget and the evaluated models and monitor rule.

VI. DISCUSSION

A. Interpretation of the Main Findings

The study asked whether a value estimator fitted on clean data helps allocate a limited monitoring budget under distribution shift, compared with simpler selection policies. In the evaluated setting the answer is no: the estimator did not outperform confidence-based selection, and the decision rule returned that verdict for the single seed run.

The ranking analysis shows a related pattern. The gate's Spearman correlation with V^* was positive at all five severities (0.109 to 0.138), but the baselines reached about 0.216 on average. Correlation is a weak summary here because V^* is mostly zero with heavy ties, and a positive correlation alone is

TABLE V

POLICY PERFORMANCE AT THE 20% BUDGET, MACRO-AVERAGED OVER FIVE GAUSSIAN-NOISE CONDITIONS (SEED 42). THE LAST TWO ROWS USE THRESHOLDS CALIBRATED ON CLEAN DATA AND ARE NOT BUDGET-MATCHED.

Policy	Esc. rate	Utility	Gain	Error (served)	Coverage	Calls/img	Spearman
Cheap-only	0.000	0.1620	0.0000	0.4190	1.0000	0.00	—
Random (200 draws)	0.200	0.1793	0.0173	0.4085	0.9519	0.60	—
Low confidence	0.200	0.2168	0.0548	0.3799	0.8857	0.60	0.2162
High entropy	0.200	0.2154	0.0534	0.3808	0.8858	0.60	0.2161
Low margin	0.200	0.2156	0.0536	0.3806	0.8866	0.60	0.2137
Value gate (entropy-only)	0.200	0.2069	0.0449	0.3882	0.9001	0.60	0.1094
Value gate (3 features)	0.200	0.2075	0.0455	0.3875	0.8981	0.60	0.1278
Direct abstention (matched)	0.000	0.2046	0.0426	0.3891	0.8981	0.00	—
Oracle (reference)	0.200	0.3229	0.1609	0.3168	0.8391	0.60	—
Always-on	1.000	0.2481	0.0861	0.3546	0.7594	3.00	—
Low confidence, fixed thr.	0.608	0.2491	0.0871	0.3555	0.7701	1.82	—
Value gate, fixed thr.	0.783	0.2294	0.0674	0.3726	0.8211	2.35	—

TABLE VI

FEATURE ABLATION AT THE 20% BUDGET (MEAN OVER FIVE GAUSSIAN-NOISE CONDITIONS).

Estimator	Mean utility gain	Mean Spearman
Entropy only	0.0449	0.1094
Confidence, entropy, margin	0.0455	0.1278

TABLE VII

BATCH RUNTIME MEASUREMENTS (2,000 IMAGES, ONE GPU, BATCH SIZE 256, SEVEN REPETITIONS). P95 IS THE 95TH PERCENTILE OF BATCH EXECUTION TIME OVER THE SEVEN REPETITIONS.

Component or pipeline	Mean (s)	P95 (s)	ms/image
ResNet-18 (base)	0.2964	0.2994	0.1482
ResNet-34	0.5108	0.5138	0.2554
ResNet-50	0.6849	0.6895	0.3424
DenseNet-121	0.6375	0.6419	0.3187
Random Forest gate	0.0694	0.0759	0.0347
Pipeline: cheap-only	0.3018	0.3047	0.1509
Pipeline: selective (20%)	0.7523	0.7606	0.3761
Pipeline: always-on	2.2048	2.2188	1.1024

not evidence of practical benefit. The utility comparison agrees with the correlation, however: the uncertainty rules selected more beneficial inputs.

One possible explanation is how little signal the fitting data contain. On the clean gate-fit split only 70 of 1,500 targets were non-zero, because the base classifier is wrong on only about 7% of clean images. A forest fitted on such sparse targets may learn a smoothed ranking that does not reflect the much higher error rate under noise (41.9% on average). Another possibility is that the relation between the cheap features and monitor benefit changes under noise, so the clean-fitted mapping transfers poorly. We did not test either explanation. Testing them would need, for example, fitting the gate on mild corruption, varying the size of the fitting set, or comparing feature distributions between clean and corrupted data.

The oracle result frames the size of the opportunity, although the oracle uses ground-truth information that is unavailable at deployment. At the same 20% budget the oracle gained 0.1609, nearly three times the best deployable rule. The three cheap features therefore carry limited information about which inputs the monitor will actually help. The oracle also beat always-on monitoring, which shows that escalating every input is not optimal under this utility because the monitor sometimes withholds correct answers.

The ablation shows that adding confidence and margin to entropy changed utility by only 0.0006. This is consistent with the three features being strongly related, since all are functions of the same probability vector. No interval is available for this small difference, so it should not be read as a meaningful improvement.

B. Relationship to Direct Abstention and Always-On Monitoring

The value gate's mean utility was 0.0029 higher than matched direct abstention at the same coverage, and it used 0.6 monitor calls per image while direct abstention used none. The difference is small and has no interval, so we do not claim that the gate beats direct abstention. Direct abstention needs no second model, which makes it a strong baseline when the monitor is expensive.

Relative to always-on monitoring, selective policies kept a substantial share of the gain at 20% of the monitor calls: about 53% for the gate and about 64% for low confidence. This supports selective escalation of this monitor in the tested setting, though not the specific value-based ranking.

C. Behaviour Under Distribution Shift

Within Gaussian noise, the gate's ranking correlation did not fall as severity increased. We cannot say whether this holds for other corruption families, since blur, frost and JPEG compression were not evaluated.

The fixed-threshold results show a separate practical issue. A threshold calibrated on clean data to escalate 20% of inputs escalated 61% to 78% of shifted inputs on average. The confidence rule's rate rose with severity because confidence dropped, whereas the gate's rate was already high at severity 1 and decreased. A fixed-budget batch rule avoids this drift but requires processing inputs in batches. A threshold rule needs

periodic recalibration or a different control mechanism under shift.

D. Practical Trade-offs

The gate costs 0.035 ms per image on this hardware, about 23% of the base model's time, and the selective pipeline cost about one third of always-on. If the gate were to match or beat confidence, this overhead would need to be weighed against a rule that costs nothing beyond the base forward pass. Since low confidence performed better in utility and requires no fitted model, it is the preferable choice in this setting.

E. Limitations

Scope. Only Gaussian noise at five severities, one split seed and the 20% budget were evaluated. The conclusions do not extend to defocus blur, frost, JPEG compression or other shifts, and consistency across split seeds and budgets could not be assessed.

Pretrained models and split design. The pretrained networks saw all 50,000 training images. The gate, calibration and evaluation sets are all drawn from the official CIFAR-10 test set. The three subsets are disjoint, but the official test set is therefore not reserved exclusively for final evaluation, and the results are not equivalent to a protocol that fits the gate on an independent labeled holdout. A stronger design would fit and calibrate on such a holdout and reserve the official test set and CIFAR-10-C for evaluation.

Seeds. A split seed varies the gate and the partition only. It does not capture variability from training the networks.

Monitor and models. The 2-of-3 acceptance rule was not tuned. The monitors use heterogeneous architectures, and different ensembles may change the relationship between cheap features and monitor benefit.

Gate fitting. The gate was fitted on 1,500 clean images with 70 non-zero targets. A larger or more informative fitting set might change the result.

Features and utility. The estimator uses three inexpensive features derived from one probability vector. The $+1/-1/0$ utility is one choice; other costs of errors and abstentions could change which policy is preferable.

Statistics and batching. The bootstrap is conditional on the fitted gate and holds selections fixed. No hypothesis-test p-value is reported. The gate-versus-direct-abstention and ablation comparisons lack intervals. Ranking correlations in Table III and severity-level threshold rates were read from figures. The top-k policy operates on a batch and is not a streaming rule. Latency is measured at component level on one machine, the P95 values come from only seven repetitions of batch execution, and none of this is a production estimate.

Taken together, these limitations mean the study establishes the behaviour of the evaluated policies under this narrow setting, but not general reliability across models, datasets or shifts.

VII. Conclusion

This study investigated value-based escalation for selective reliability monitoring under distribution shift. A Random Forest estimator, fitted on clean CIFAR-10 using confidence, entropy and margin, was used to allocate a disagreement-based monitor under a fixed budget and was evaluated against uncertainty-based, random, cheap-only, always-on and direct-abstention policies on CIFAR-10-C Gaussian noise.

In this pilot-scale evaluation, the estimator did not outperform simple confidence-based selection. At the 20% budget its mean utility gain was 0.0455 against 0.0548 for low confidence, a paired difference of -0.0094 (95% CI -0.0120 to -0.0069) for one split seed. An entropy-only estimator performed almost identically, and the oracle's gain of 0.1609 shows that the cheap features captured only a small part of the possible benefit. Selective monitoring at 20% cost about one third of always-on batch latency while retaining roughly half to two thirds of its utility gain. Thresholds calibrated on clean data did not hold the intended escalation rate under shift.

The conclusions are limited to the evaluated pretrained models, the 2-of-3 monitor rule, three cheap features, one corruption family, one split seed, one primary budget and the stated utility. Future work should evaluate the remaining corruption families, additional split seeds and other budgets, fit the estimator on an independent holdout, examine richer features and larger fitting sets, and test whether the gate's disadvantage persists.

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